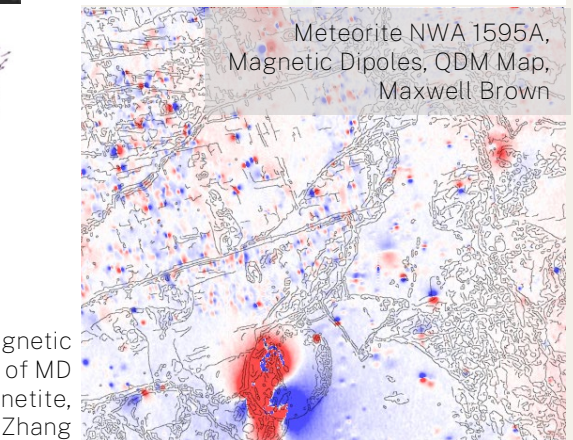
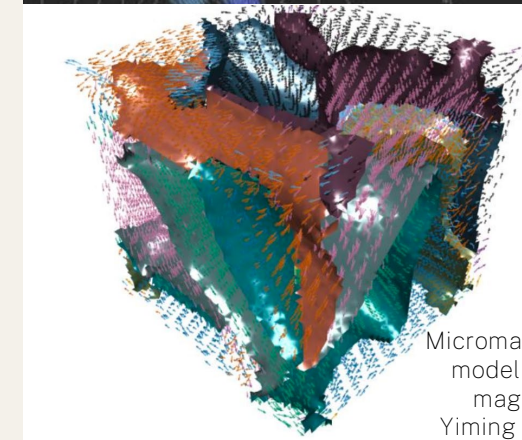
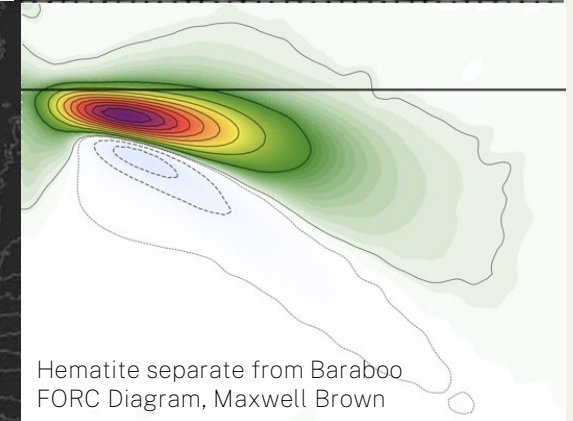
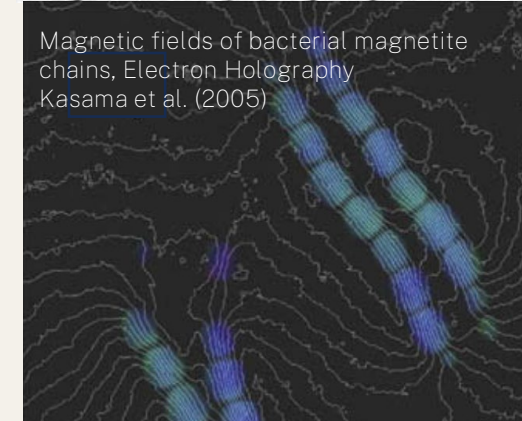
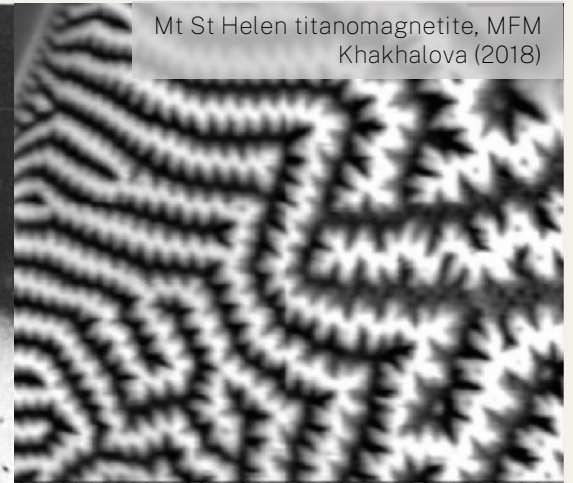
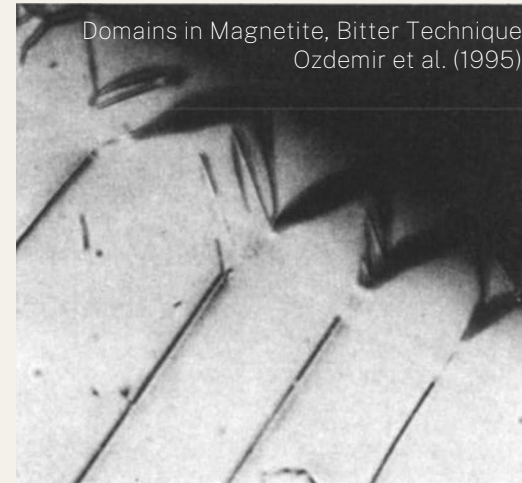


Overview – Useful Concepts - Applications

Introduction to Rock Magnetism

IRM Summer School 2026
Day 1

Monday July 20th, 2026, Maxwell Brown



Who I am?

- Maxwell Brown (Max)
- SSRM lecturer; susceptibility/AMS measurements.
- Logistics – here to help you with organizational questions.
- Facility Manager of the IRM since 2020.
- Background in Geophysics.
- PhD on geomagnetic field reversals.
- Interests:
 - Long term geomagnetic field variations
 - Archaeomagnetism
 - Magnetic mineralogy
 - QDM
 - Databases and analysis software.



We'll be going up here on the weekend, smoke depending

What is the “rock” in rock magnetism?

“Rock” means any material containing magnetic minerals — from lava and mud to meteorites, bacteria, pollution and brain tissue.



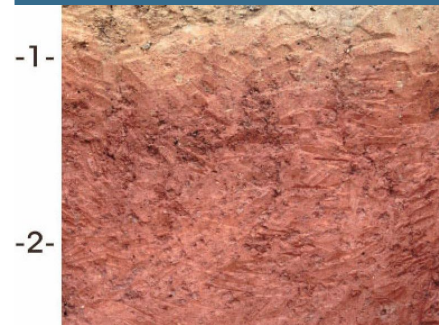
01 Extrusive and intrusive igneous rocks



02 Sedimentary rock



03 Marine & lake sediments



04 Soils



05 Archaeological artifacts



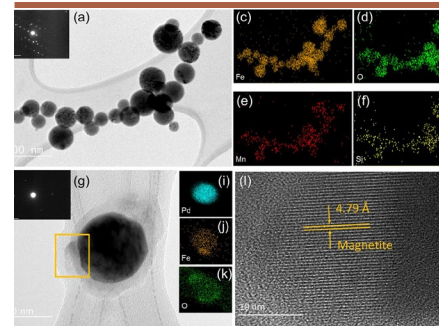
06 Meteorites



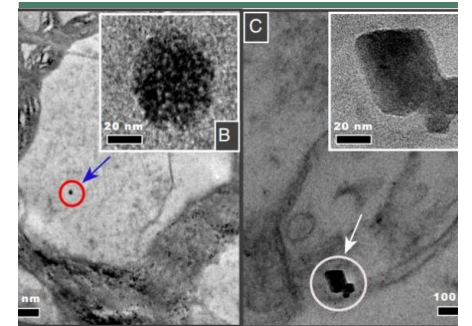
07 Moon and Mars rocks



08 Magnetotactic bacteria



09 Magnetic pollution

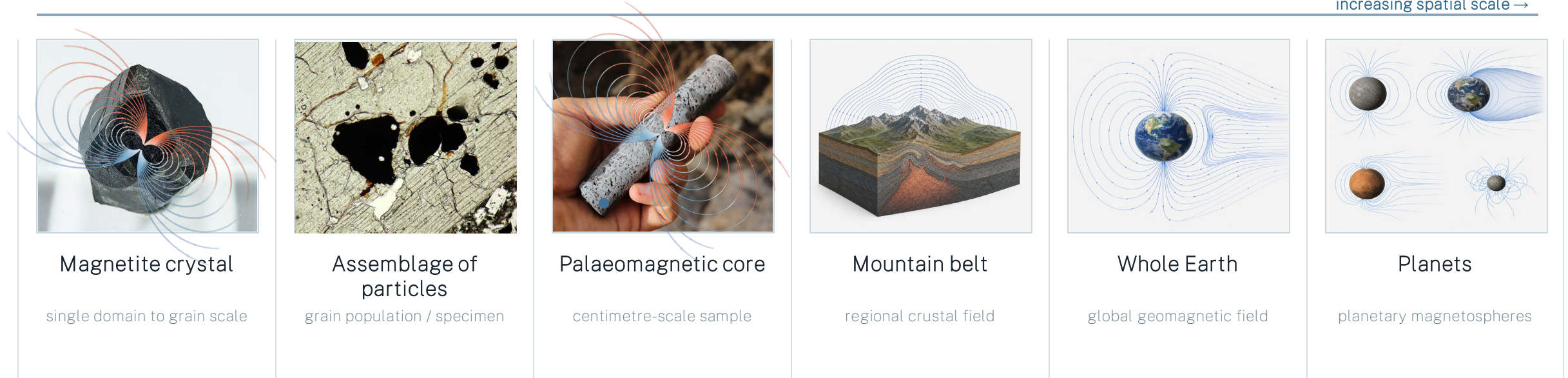


10 Brain magnetite

What is the “magnetism” in rock magnetism?

Magnetic fields occur from individual crystals to entire planets (and across the universe)

increasing spatial scale →



HOW THE FIELDS CAN BE OBSERVED

CRYSTAL Magnetic force microscopy (MFM)	PARTICLE ASSEMBLAGE QDM / SQUID microscopy Laboratory magnetic techniques	PMAG CORE SQUID magnetometers Remanence and demagnetisation measurements	MOUNTAIN BELT Aeromagnetic surveys Regional magnetic anomaly mapping	WHOLE EARTH Satellites; ground observatories Archaeological materials; paleomagnetic rock samples; sediment cubes	PLANETS Satellite missions Orbital and fly-by magnetometer observations
--	---	---	--	---	--

Early Observations of Magnetism

Thales of Miletus
Greek Philosopher, 6th Century B.C.



<https://www.britannica.com/biography/Thales-of-Miletus#/media/1/589798/94991>

Lucretius,
Roman Philosopher, 1st Century B.C.



<https://archive.org/details/tlucretiuscaruso01lucr/page/n5/mode/2up>

Key works in rock magnetism

1938



Johann
Königsberger

Established how natural remanence can dominate a rock's induced magnetization.

Natural remanence in eruptive rocks
 Q = remanent / induced magnetization

DEFINING WORK

Natural remanent magnetization of eruptive rocks

Legacy: the Königsberger ratio, Q .

1938–59



Émile
Thellier

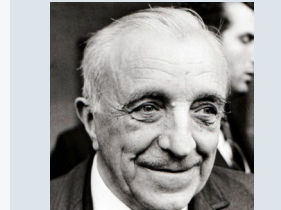
Laid out principles of Thermoremanent magnetization (TRM). Developed the laws of partial thermoremanent magnetization (pTRM)

Pioneer of paleointensity with Odete Thellier, with their work on archaeological materials.

DEFINING WORKS

1938 PhD Thesis & 1959 paper on paleointensity

1949



Louis
Néel (Nobel Prize)

Theoretical work on ferri and antiferrimagnets. Explained stable remanence and thermal relaxation in fine, single-domain magnetic grains, building upon the observation principals of Thellier.

DEFINING WORK

Theory of fine-grained ferromagnets

Foundation for blocking temperature and Néel relaxation.

1948



Stoner &
Wohlfarth

Modeled coherent rotation and hysteresis in assemblies of non-interacting single-domain particles – still our best model for SD hysteresis.

DEFINING WORK

“A mechanism of magnetic hysteresis...”

The Stoner–Wohlfarth model.

1953–61



Takeshi
Nagata

Unified experimental mineral magnetism, remanence, and geophysical applications.

DEFINING WORK

Rock Magnetism

The first major textbook synthesis of the field.

1974



Stacey &
Banerjee

Connected magnetic theory with the behavior of real rocks and magnetic minerals.

Domains • PSD • coercivity remanence • mineral physics

DEFINING WORK

The Physical Principles of Rock Magnetism

A rigorous physics framework for the discipline.

1997



Dunlop &
Özdemir

Integrated decades of theory, experiments, mineralogy, and environmental applications.

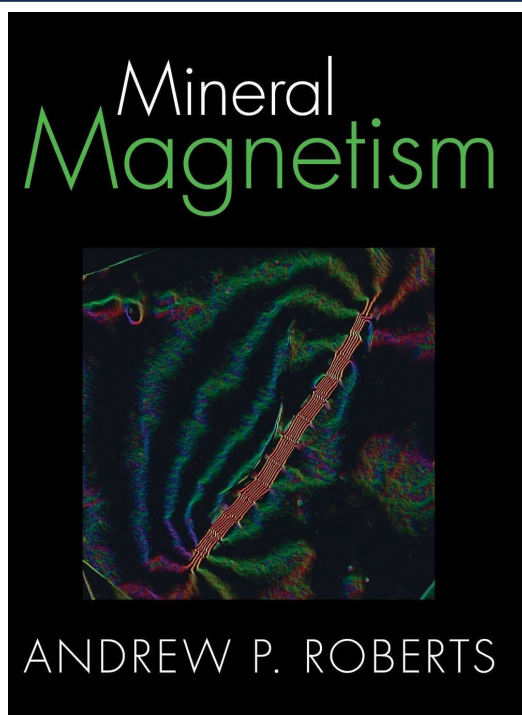
DEFINING WORK

Rock Magnetism: Fundamentals and Frontiers

The modern comprehensive reference.



Modern Texts

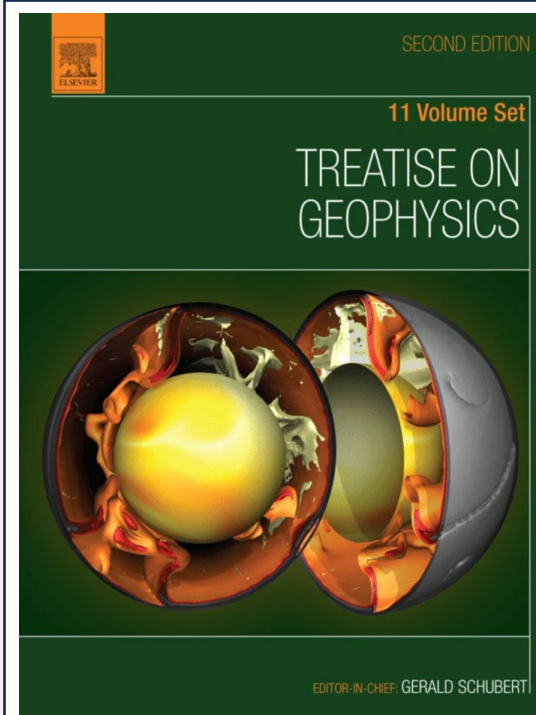


Part I Essentials of Mineral Magnetism

1	Introduction	3
2	Fundamentals of Mineralogy	11
3	Fundamentals of Mineral Magnetism	31
4	Experimental Methods in Mineral Magnetism	72

Part II Magnetism of the Major Terrestrial Rock-Forming Magnetic Minerals

5	The Iron–Titanium Oxides: Magnetite (Fe_3O_4)	105
6	The Iron–Titanium Oxides: The Titanomagnetite ($\text{Fe}_{3-x}\text{Ti}_x\text{O}_4$) Series	142
7	The Iron–Titanium Oxides: Hematite ($\alpha\text{-Fe}_2\text{O}_3$)	182
8	The Iron–Titanium Oxides: The Titanohematite ($\text{Fe}_{2-y}\text{Ti}_y\text{O}_3$) Series	229
9	The Iron–Titanium Oxides: Aluminium Hematite ($\alpha\text{-Fe}_{2-y}\text{Al}_y\text{O}_3$)	267
10	The Iron–Titanium Oxides: Maghemite ($\gamma\text{-Fe}_2\text{O}_3$) and Titanomaghemite	284
11	The Iron Oxyhydroxides: Goethite ($\alpha\text{-FeOOH}$)	322
12	The Iron Sulphides: Pyrrhotite (Fe_{1-x}S)	356
13	The Iron Sulphides: Greigite (Fe_3S_4)	392
14	Miscellaneous: Other Magnetic Minerals	417



2.24. Magnetic Properties of Rocks and Minerals

RJ Harrison, RE Dunin-Borkowski, T Kasama, ET Simpson, JM Feinberg

5.08. Magnetizations in Rocks and Minerals

DJ Dunlop and O Ozdemir

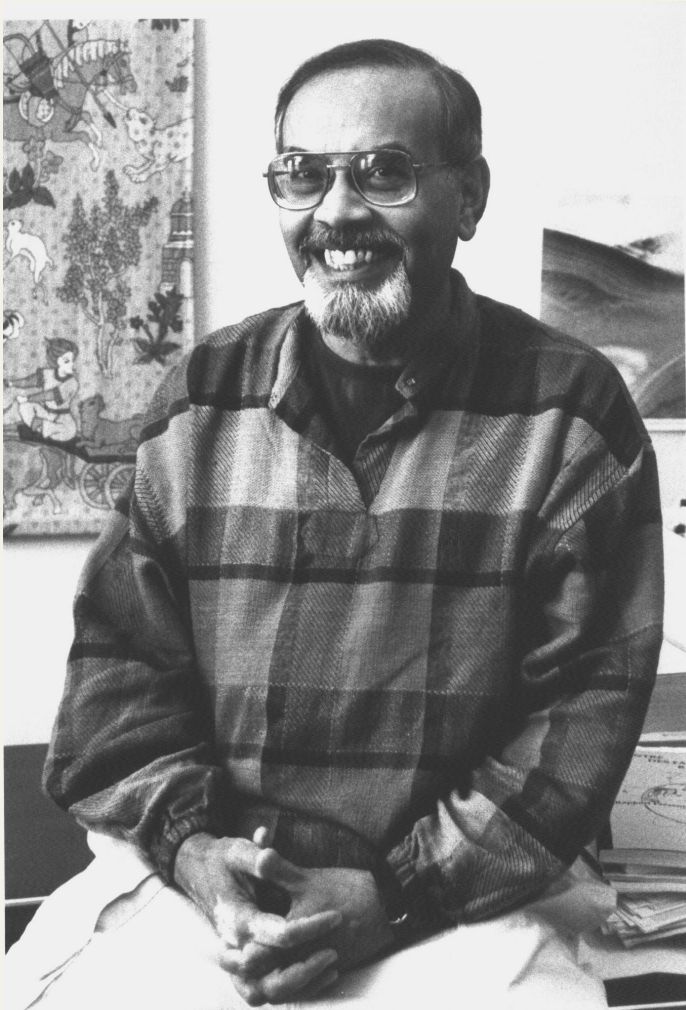
11.05. Geophysical Properties of the Near-Surface Earth: Magnetic Properties

BM Moskowitz, M Jackson, V Chandler

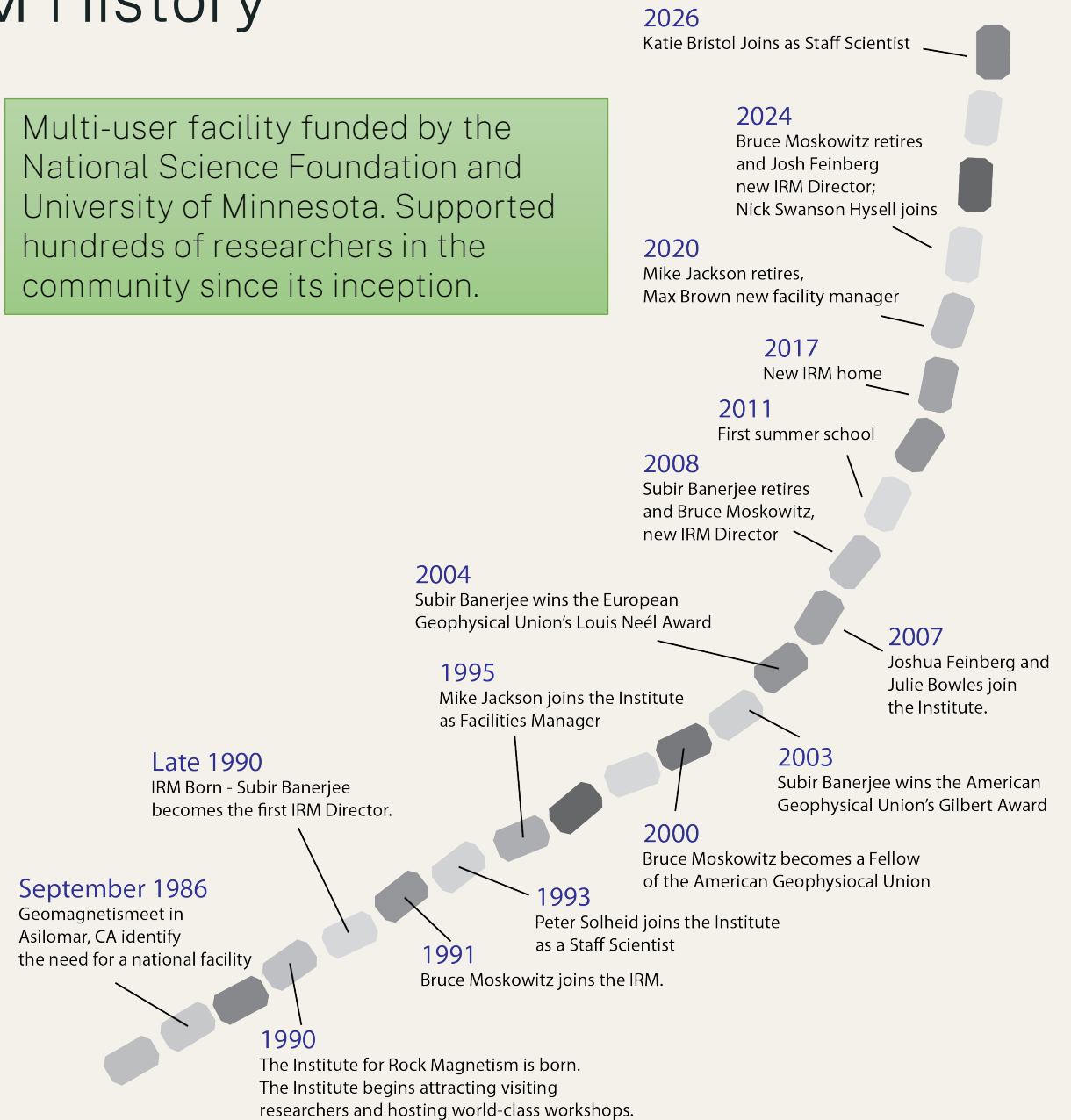
You can find these in the student material google drive

IRM History

IRM Founded in 1990 by Subir Banerjee



Multi-user facility funded by the National Science Foundation and University of Minnesota. Supported hundreds of researchers in the community since its inception.



Where does magnetism come from?

Magnetism is associated with only a handful of elements (Fe, Ni, Co, Cr, Mn and some rare earths), of which iron is by far the most common in geological materials.

Rock magnetism, in a broad sense, is the means of studying the geology, geochemistry and geophysics of iron

M. Jackson

Magnetic Periodic Table

Atomic Number →

Typical ionic charge →

Antiferromagnetic T_N (K) →

66 **Dy**

162.5

3 + 4f

179 85

← Atomic symbol

← Atomic weight

← Ferromagnetic T_C (K)

1 H 1.00																		2 He 4.00																	
3 Li 6.94 1 + 2s ⁰		4 Be 9.01 2 + 2s ⁰																5 B 10.81		6 C 12.01		7 N 14.01		8 O 16.00 35		9 F 19.00		10 Ne 20.18							
11 Na 22.99 1 + 3s ⁰		12 Mg 24.21 2 + 3s ⁰																13 Al 26.98 3 + 2p ⁶		14 Si 28.09		15 P 30.97		16 S 32.07		17 Cl 35.45		18 Ar 39.95							
19 K 38.21 1 + 4s ⁰		20 Ca 40.08 2 + 4s ⁰		21 Sc 44.96 3 + 3d ⁰		22 Ti 47.88 4 + 3d ⁰		23 V 50.94 3 + 3d ²		24 Cr 52.00 3 + 3d ³ 312		25 Mn 55.85 2 + 3d ⁵ 96		26 Fe 55.85 3 + 3d ⁶ 1043		27 Co 58.93 2 + 3d ⁷ 1390		28 Ni 58.69 2 + 3d ⁸ 629		29 Cu 63.55 2 + 3d ⁹		30 Zn 65.39 2 + 3d ¹⁰		31 Ga 69.72 3 + 3d ¹⁰		32 Ge 72.61		33 As 74.92		34 Se 78.96		35 Br 79.90		36 Kr 83.80	
37 Rb 85.47 1 + 5s ⁰		38 Sr 87.62 2 + 5s ⁰		39 Y 88.91 2 + 4d ⁰		40 Zr 91.22 4 + 4d ⁰		41 Nb 92.91 5 + 4d ⁰		42 Mo 95.94 5 + 4d ¹		43 Tc 97.9		44 Ru 101.1 3 + 4d ⁶		45 Rh 102.4 3 + 4d ⁶		46 Pd 106.4 2 + 4d ⁸		47 Ag 107.9 1 + 4d ¹⁰		48 Cd 112.4 2 + 4d ¹⁰		49 In 114.8 3 + 4d ¹⁰		50 Sn 118.7 4 + 4d ¹⁰		51 Sb 121.8		52 Te 127.6		53 I 126.9		54 Xe 83.80	
55 Cs 132.9 1 + 6s ⁰		56 Ba 137.3 2 + 6s ⁰		57 La 138.9 3 + 4f ⁰		72 Hf 178.5 4 + 5d ⁰		73 Ta 180.9 5 + 5d ⁰		74 W 183.8 6 + 5d ⁰		75 Re 186.2 4 + 5d ⁶		76 Os 190.2 3 + 5d ⁶		77 Ir 192.2 4 + 5d ⁶		78 Pt 195.1 2 + 5d ⁸		79 Au 197.0 1 + 5d ¹⁰		80 Hg 200.6 2 + 5d ¹⁰		81 Tl 204.4 3 + 5d ¹⁰		82 Pb 207.2 4 + 5d ¹⁰		83 Bi 209.0		84 Po 209		85 At 210		86 Rn 222	
87 Fr 223		88 Ra 226.0 2 + 7s ⁰		89 Ac 227.0 3 + 5f ⁰																															
				58 Ce 140.1 4 + 4f ⁰ 13		59 Pr 140.9 3 + 4f ¹		60 Nd 144.2 3 + 4f ² 19		61 Pm 145		62 Sm 150.4 3 + 4f ³ 105		63 Eu 152.0 2 + 4f ⁴ 90		64 Gd 157.3 3 + 4f ⁵ 292		65 Tb 158.9 3 + 4f ⁶ 229 221		66 Dy 162.5 3 + 4f ⁷ 179 85		67 Ho 164.9 3 + 4f ⁸ 132 20		68 Er 167.3 3 + 4f ⁹ 85 20		69 Tm 168.9 3 + 4f ¹⁰ 56		70 Yb 173.0 3 + 4f ¹⁰		71 Lu 175.0 3 + 4f ¹⁴					
				90 Th 232.0 4 + 5f ⁰		91 Pa 231.0 5 + 5f ⁰		92 U 238.0 4 + 5f ²		93 Np 238.0 5 + 5f ²		94 Pu 244		95 Am 243		96 Cm 247		97 Bk 247		98 Cf 251		99 Es 252		100 Fm 267		101 Md 268		102 No 269		103 Lr 260					

Nonmetal

Metal

Radioactive

Diamagnet

Paramagnet

BOLD

 Magnetic atom

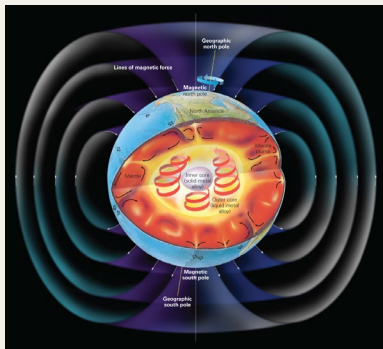
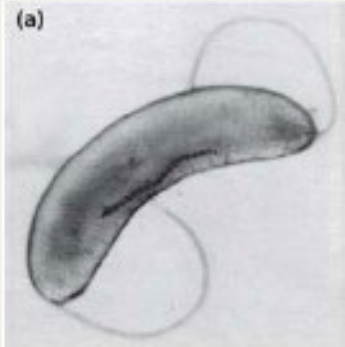
Ferromagnet $T_C > 290K$

Antiferromagnet with $T_N > 290K$

Antiferromagnet/Ferromagnet with $T_N/T_C < 290 K$

What is a Magnet?

Let's ask ChatGPT



What is the simplest description of a magnetic field?

A magnetic field is the invisible influence around a magnet that can push, pull, or turn magnetic objects.



What is a magnetic object?

A **magnetic object** is something that either acts like a magnet or responds to one—for example, a magnet, a piece of iron, or a compass needle.



Magnetism as Poles

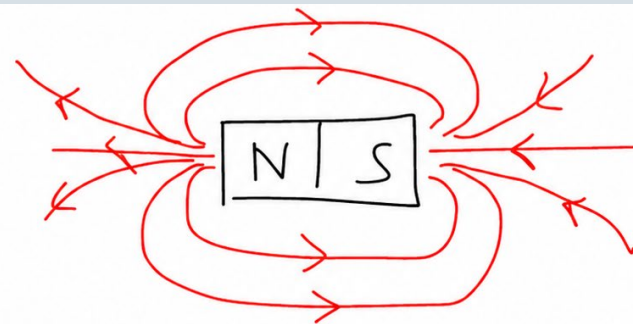
Equal Poles Repel



Unequal Poles Attract



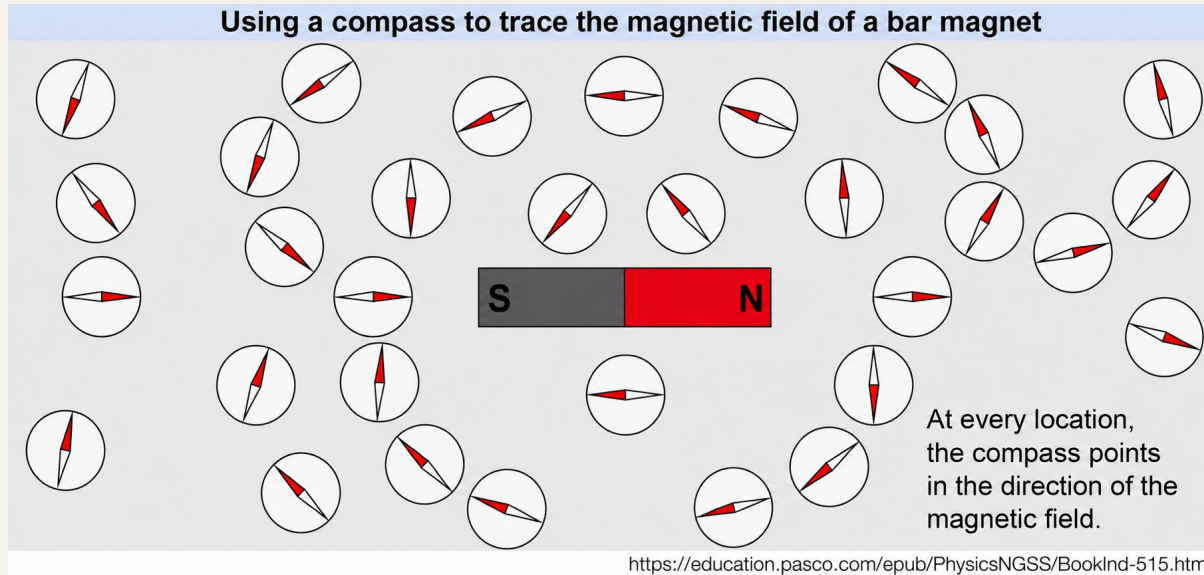
Lines of Magnetic Field



Magnets have a North and a South pole

Bruce Moskowitz

What is a magnetic field?



An invisible vector field that interacts with other **magnets**.

Defined not by its cause but by forces it exert on physical objects.

Magnetic fields

- Exert a torque on a compass tending to align their axes with the field.
- Exert a force on a moving charged particle, either in free space or through a conductor as current.

How to generate a magnetic field

- Electromagnets → requires electrical energy (currents).
- Permanent magnets → assembly of dipoles in materials made of magnetic material.

Magnetic B-Field

Coulomb's Law for magnetic poles

Magnetic Force:

$$F_M = K \frac{p_1 p_2}{r^2}$$

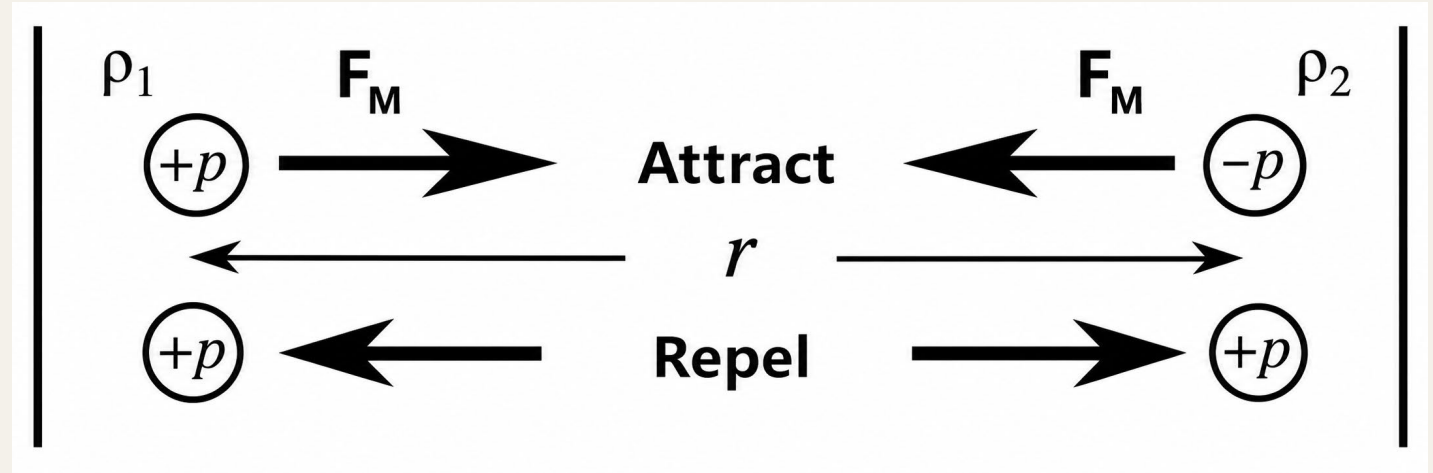
Magnetic induction/flux density:

$$B(r) = \frac{F_M}{\rho_1} = \frac{\mu_0}{4\pi} \frac{\rho_2}{r^2}$$

$$K = \frac{\mu_0}{4\pi} \text{ (SI units)}$$

Permeability of free space:

$$\mu_0 = 4\pi \cdot 10^{-7} \text{ mkg s}^{-2} \text{ A}^{-2}$$



$+p$, north pole with strength $\rho_1 = p_1$

$-p$, south pole with strength $\rho_2 = p_2$

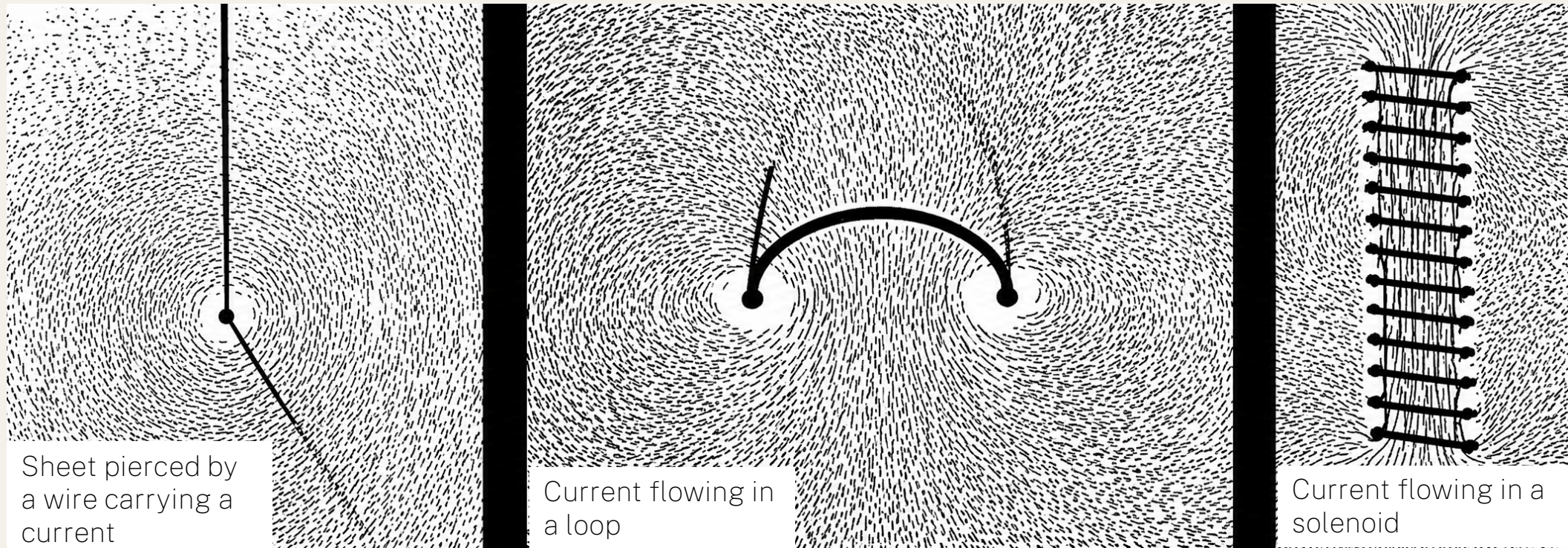
r , distance between poles

$$B \text{ in Tesla or } \frac{\text{Weber}}{\text{m}^2}$$

Magnetic B-Field

Distribution of iron filings on a flat sheet
Magnetic field patterns for various current
distributions

After Jiles, 1991



Magnetic materials produce similar effects to ones created by electric circuits

All magnetism can be attributed to *electric currents* or electron spin.

There are *no magnetic currents*.

A magnetized particle is a dipole: magnetic flux always closes

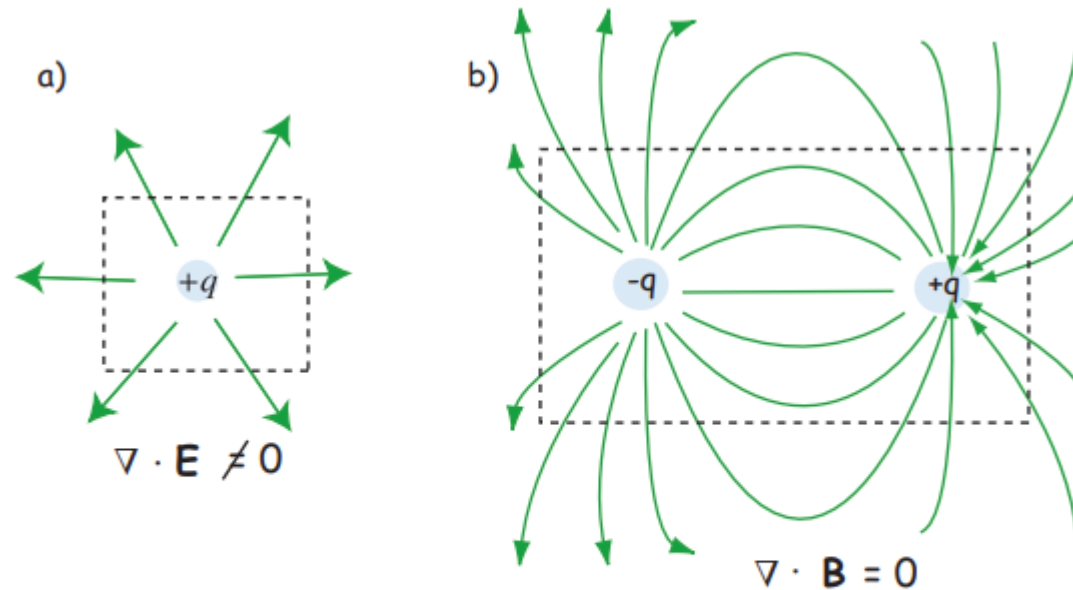


FIGURE 1.6. a) An electric charge produces a field that diverges out from the source. There is a net flux out of the dashed box, quantified by the divergence ($\nabla \cdot \mathbf{E}$), which is proportional to the magnitude of the sources inside the box. b) There are no isolated magnetic charges, only dipoles. Within any space (e.g., the dashed box), any flux line that goes out must return. The divergence of such a field is zero, i.e., $\nabla \cdot \mathbf{B} = 0$.

Tauxe, Essentials

WHY MAGNETIC FIELDS DIFFER

1 No isolated magnetic pole

Unlike an electric charge $+q$ or $-q$, a north or south magnetic pole has not been observed by itself.

2 Magnetism appears as dipoles

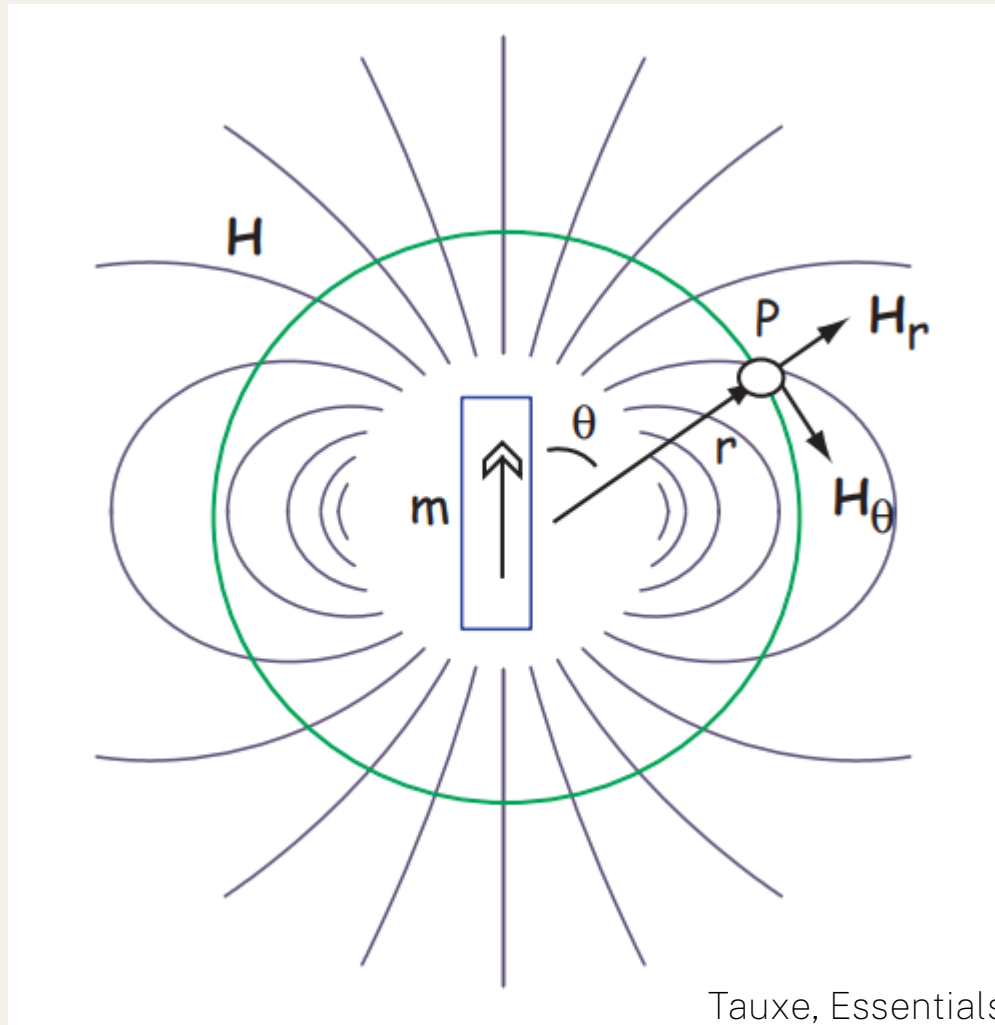
A magnetized particle carries paired N and S poles. Outside the particle, $\mathbf{B}$ runs from N to S; inside, it returns from S to N.

3 Field lines form closed loops

No field line begins or ends inside the box, so outward and inward magnetic flux cancel exactly.

Every magnetic field line that leaves one pole returns to the opposite pole — so a closed surface encloses zero net magnetic flux.

Away from the source, the field looks like a dipole



FROM SOURCE TO OBSERVATION

1 Potential away from the source

$$\nabla^2 \psi_m = 0$$

$$\psi_m = mr / (4\pi r^2) = m \cos \theta / (4\pi r^2)$$

2 Take the spatial gradient

$$\mathbf{H} = -\nabla \psi_m$$

$$H_r = 2m \cos \theta / (4\pi r^3)$$

$$H_\theta = m \sin \theta / (4\pi r^3)$$

3 Convert $\mathbf{H}$ to flux density $\mathbf{B}$

$$\mathbf{B} = \mu_0 \mathbf{H} \quad (\text{air / free space})$$

$$|\mathbf{B}| = \mu_0 m \sqrt{(1 + 3 \cos^2 \theta)} / (4\pi r^3)$$

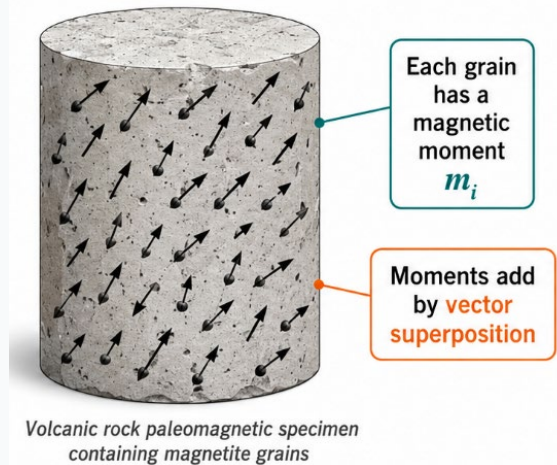
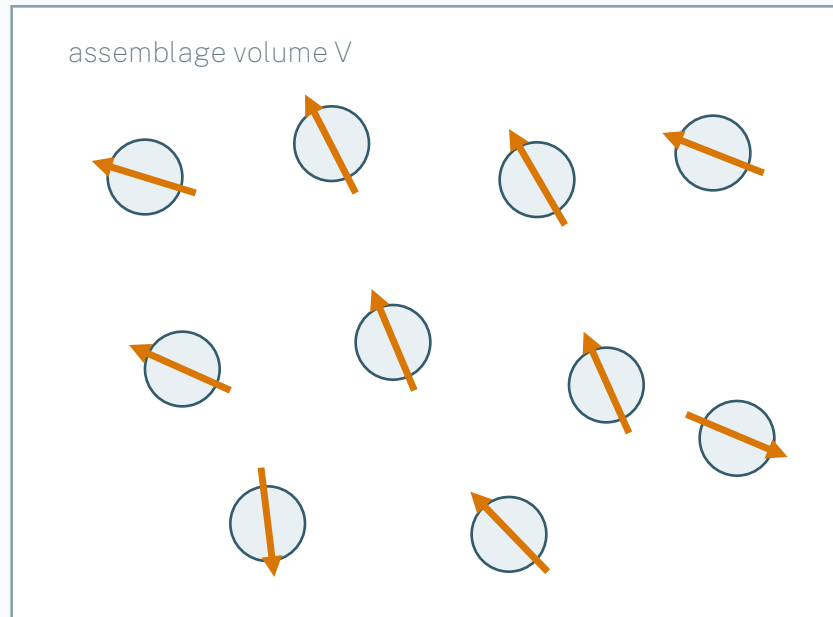
4 Project $\mathbf{B}$ onto the sensor axis

$$B_{\text{meas}} = \hat{\mathbf{s}} \cdot \mathbf{B}$$

Far from the particle, the detailed source geometry collapses to one magnetic moment m ; the measured signal decays as $1/r^3$ and depends on orientation.

An assemblage field is the vector sum of every grain dipole

FROM ONE MOMENT TO MANY



Each arrow = grain moment m_i & different r_i and directions from observation point.

SUPERPOSITION AT THE SENSOR

For grain i at position r_i :

$$R_i = r - r_i$$

$$B_i(r) = \mu_0 / (4\pi R_i^3) [3(m_i \cdot R_i)R_i - m_i]$$

Combined field:

$$B_{\text{total}}(r) = \Sigma_i B_i(r)$$

Each contribution has its own magnitude and direction because each grain has a different m_i and position r_i .

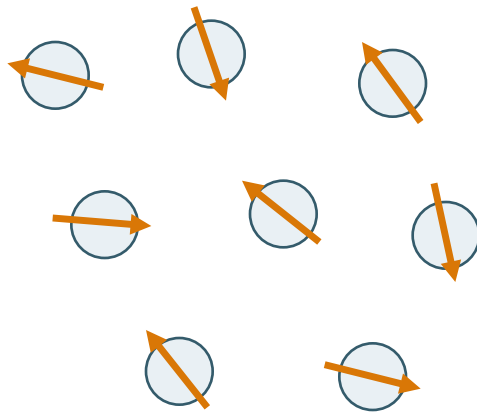
In general: $|\Sigma_i B_i| \neq \Sigma_i |B_i|$

Superposition is linear: calculate each grain's vector field at the observation point, then add the vectors — not their magnitudes.

Moment orientation determines whether grain moments add or cancel

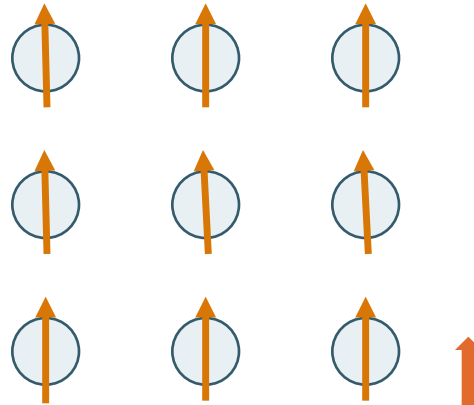
VECTOR STATISTICS OF THE ASSEMBLAGE

RANDOM ORIENTATION



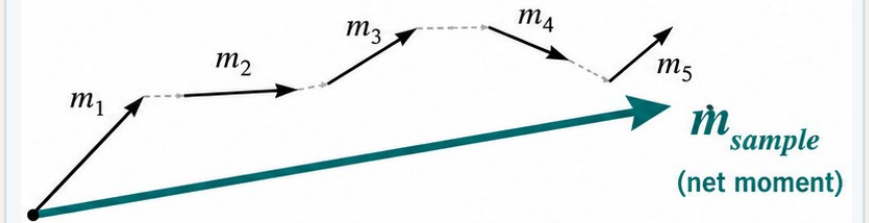
$$m_{\text{net}} \approx 0$$

ALIGNED ORIENTATION



$$m_{\text{net}} \approx N m$$

VECTOR SUM (SCHEMATIC)



1 $m_{\text{sample}} = \sum_i m_i$

2 $m_{\text{sample}} = \int_V M(r) dV$

The measured remanence is the integrated magnetization of the whole specimen volume.

Random, independent grains:

$$\langle m_{\text{net}} \rangle = 0, \text{ but } \text{rms} (|m_{\text{net}}|) \sim m\sqrt{N}$$

Fully aligned grains:

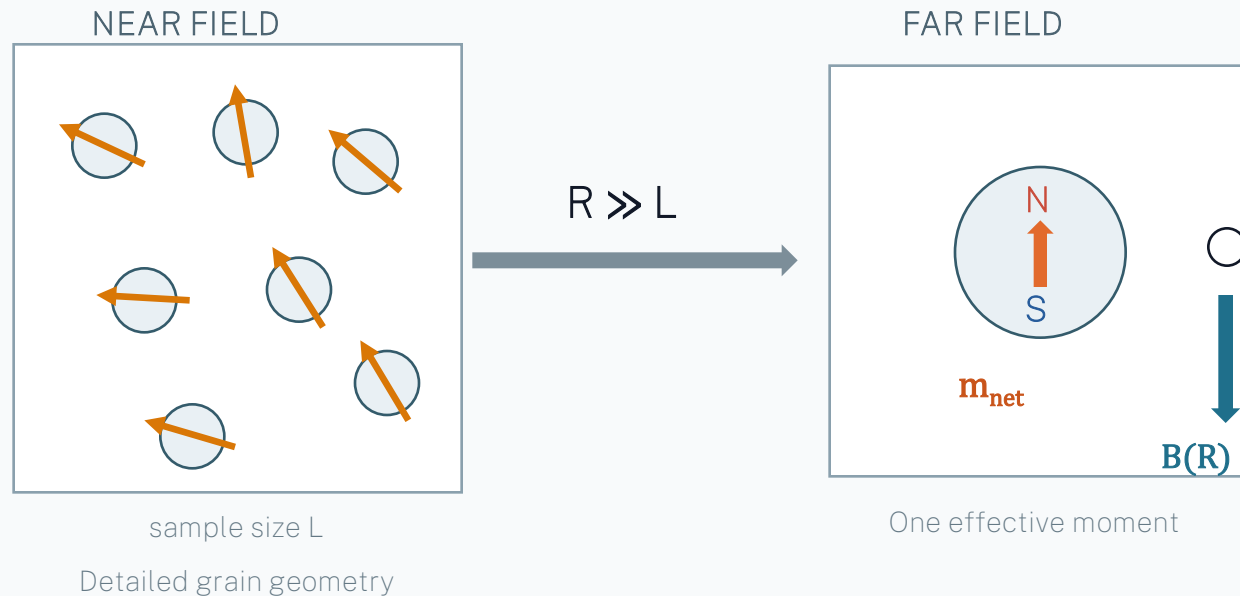
$$|m_{\text{net}}| \approx N m$$

Partial alignment lies between these limits and is controlled by magnetic history, anisotropy, temperature, and grain interactions.

A large number of magnetic grains does not guarantee a large external field to the sample: the measured moment depends on their vector alignment – permanent, remanent, or in an applied field

Far enough away, the assemblage behaves like one effective dipole

MULTIPOLE REDUCTION WITH DISTANCE



EXACT SUM VS. FAR-FIELD MODEL

Exact field at any distance

$$\mathbf{B}_i(\mathbf{r}) = \mu_0 / (4\pi R_i^3) [3(\mathbf{m}_i \cdot \mathbf{R}_i)\mathbf{R}_i - \mathbf{m}_i]$$

Effective moment

$$\mathbf{m}_{\text{net}} = \sum \mathbf{m}_i$$

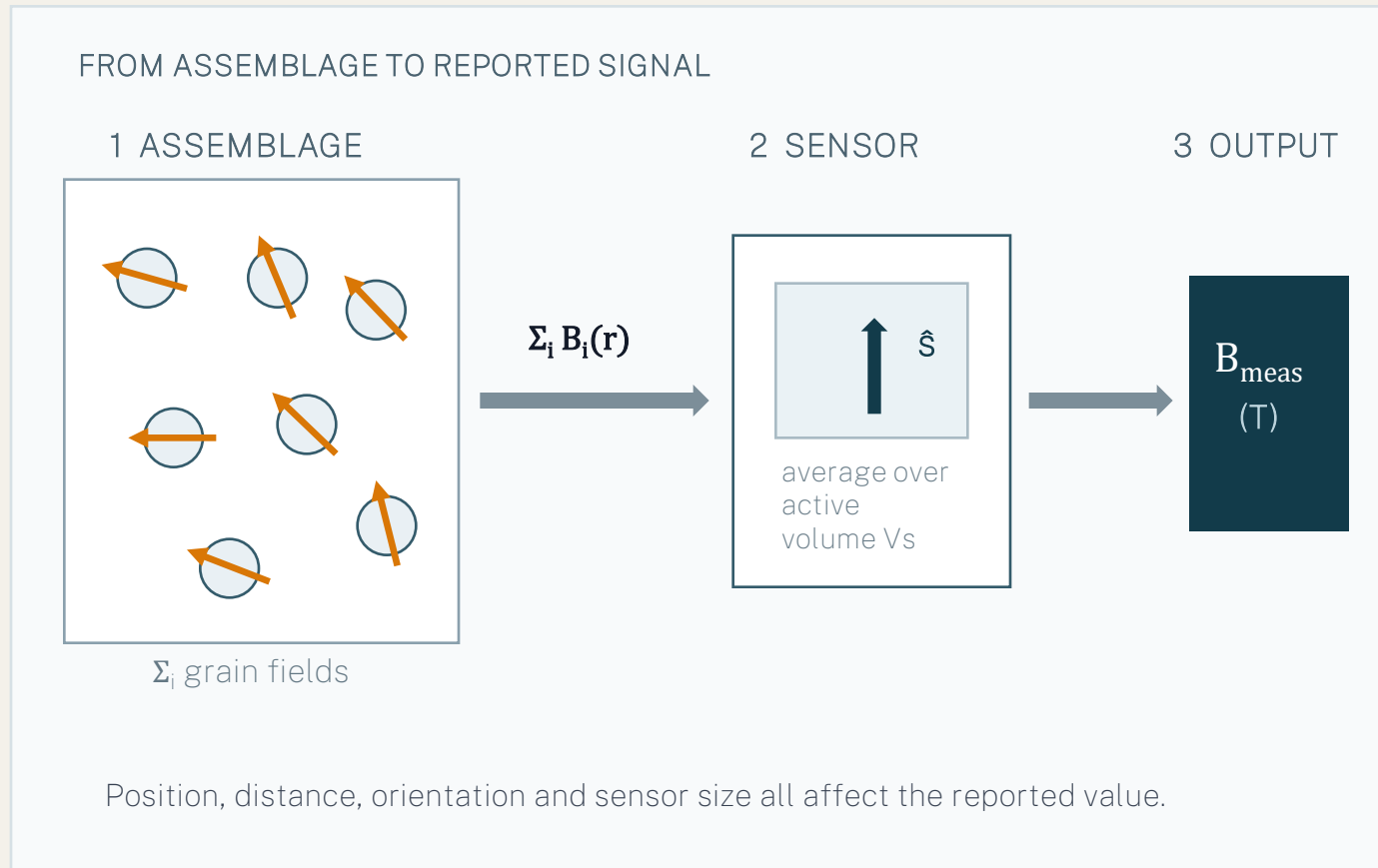
Leading far-field term

$$\mathbf{B}(\mathbf{R}) \approx \mu_0 / (4\pi R^3) \times [3(\mathbf{m}_{\text{net}} \cdot \mathbf{R})\mathbf{R} - \mathbf{m}_{\text{net}}]$$

If $\mathbf{m}_{\text{net}} \approx 0$, quadrupole and higher-order structure can dominate – but these terms decay faster than $1/R^3$.

When observation distance R is much larger than sample size L , the leading field depends only on the net moment $\mathbf{m}_{\text{net}}$ and decays as $1/R^3$.

The instrument reports a projected, spatially averaged combined field



MEASUREMENT MODEL

Point-like sensor

$$B_{\text{meas}} \approx \hat{s} \cdot B_{\text{total}}(rs)$$

Finite active volume

$$B_{\text{meas}} = \int V_s w(r) \hat{s} \cdot [\Sigma_i B_i(r)] dV$$

$w(r)$ describes sensor sensitivity and is normalized over the active volume.

Reported signal

$$S = G B_{\text{meas}} + B_{\text{background}} + \epsilon (\text{noise})$$

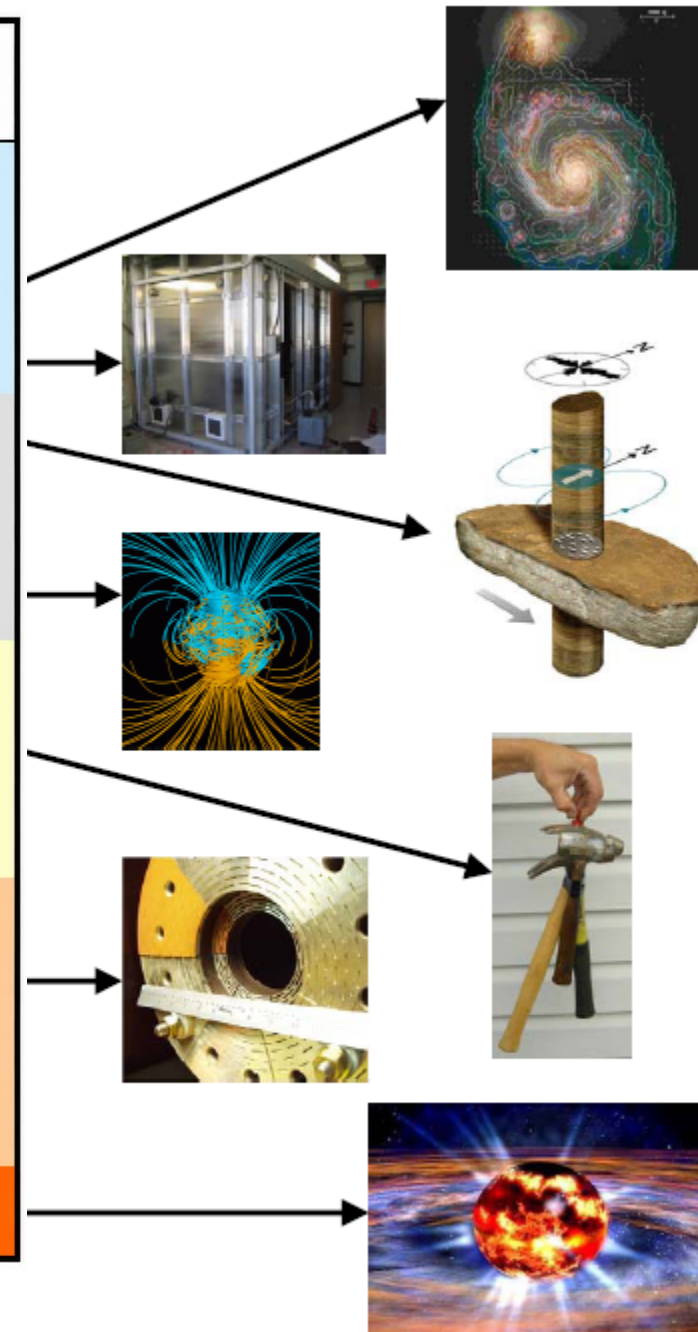
Vector magnetometer: reports B_x, B_y, B_z

Scalar magnetometer: reports $|B|$

Calibration sets gain G and sensor-axis orientation.

The measured number is not the field of one grain: it is the sensor-weighted sum of all grain fields, plus background and measurement noise.

Magnetic field source	Field Tesla
Best magnetic screen, superconducting (2008)	10^{-11}
Galactic field	10^{-9}
Good magnetic screen, conventional	10^{-9}
Field anomalies produced by crustal rocks (Earth)	10^{-7}
Strong magnetic storm (Earth)	10^{-6}
Earth magnetic field (mid latitudes)	5×10^{-5}
Commercial permanent magnet (ferrite)	0.1
Strongest permanent magnets (NdFeB, 2008)	1.4
Commercial electromagnet (FeCo poles)	2
Superconducting magnet (2008)	14
Bitter electromagnet, continuous field (2008)	45
Electromagnet, impulse field, non-explosive (2008)	100
Electromagnet, explosive field compression (2008)	600
Neutron star	10^{12}

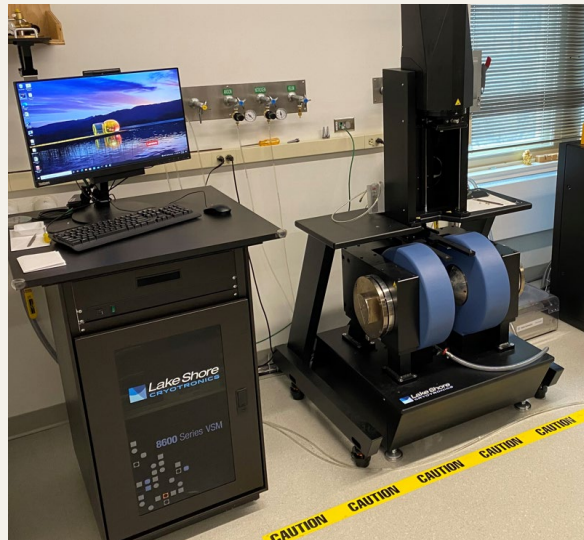


How do we measure these tiny rock fields?

Sensitive magnetometers are available to measure the weak magnetic signals in Earth (and planetary) materials



MPMS (Magnetic Property Measurement System)



Vibrating Sample magnetometer



DC SQUID U-Channel Magnetometer and Shielded Room

Each measurement opens a different window onto the same sample

FIELD-DEPENDENT

Field response

Low-field susceptibility
Hysteresis and IRM
In-field magnetization

VERY LOW → HIGH T

Temperature response

Cryogenic to high-temperature
Field-cooled / zero-field-cooled
Transitions and blocking

ZERO-FIELD

Remanence and stability

Natural and induced remanence
AF and thermal demagnetization
Relaxation and stability

WHERE + WHAT

Spatial + microstructure

Magnetic microscopy
SEM, TEM and optical microscopy
Mineralogy and chemistry

HOLISTIC SYNTHESIS

Combine observations

to resolve:

- Magnetic phases
- Grain size and domain state
- Particle interactions
- Remanence carriers
- Stability and history

Robust interpretation comes from combining field, temperature, remanence and spatial observations.

Geology, Geochemistry & Geophysics of Iron

Paleomagnetism

Study of the remanent magnetization in naturally occurring magnetic minerals.

Rock and Mineral Magnetism

Study of the physical/chemical basis of paleomagnetism and applications to geological and geophysical problems.

Environmental Magnetism

Uses identification and variations in magnetic mineralogy to track present and past changes in environment.

Rock Magnetism – How remarkable!

“It is amusing to reflect that if the pioneering paleomagnetists of the early fifties had known or even suspected, the full complexity (chemical, mineralogical, textural, magnetic-structural) of the magnetic properties of rocks, they probably would have thrown up their hands, declared rocks inherently unreliable, and turned to lesser things. Ignorance, it would seem, can sometimes be a blessing”

J. Verhoogen (1977, *Origins of Thermoremanent Magnetization*)



<http://eps.berkeley.edu/alumni/verhoogen.php>

“It is ironic that charting the movements of Earth’s tectonic plates and the evolution of the geodynamo over past eons rests ultimately on the behavior of often imperfect oxide and sulfide grains that are so small that their internal magnetic structures have, until recently, been impossible to observe.”

D. Dunlop (Phys. Today 65(6), 31 (2012); doi: 10.1063/PT.3.1604)



Annu. Rev. Earth Sci. vol 1., 1973

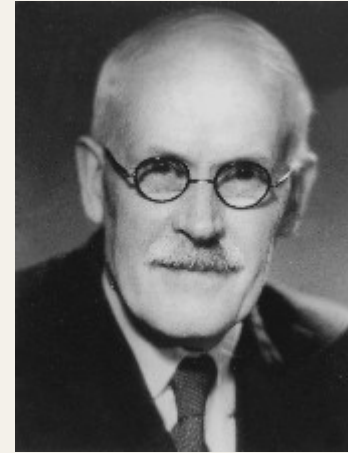
THEORY AND NATURE OF MAGNETISM IN ROCKS

R. B. Hargraves

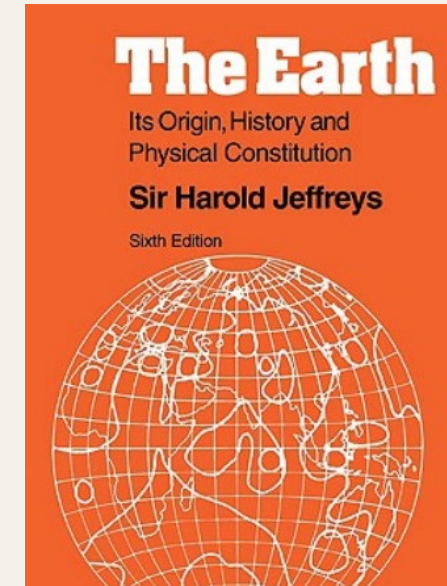
Department of Geological and Geophysical Sciences,
Princeton University, Princeton, New Jersey

S. K. Banerjee

Department of Geology and Geophysics,
University of Minnesota, Minneapolis, Minnesota



Sir Harold Jeffreys
(1891-1989)



Jeffreys, The Earth, 1970

Sir Harold Jeffreys has written (41):

When I last did a magnetic experiment (about 1909) we were warned against careless handling of permanent magnets, and the magnetism was liable to change without much carelessness. In studying the magnetism of rocks the specimen has to be broken off with a geological hammer and then carried to the laboratory. It is supposed that in the process its magnetism does not change to any important extent, and though I have often asked how this comes to be the case I have never received an answer.

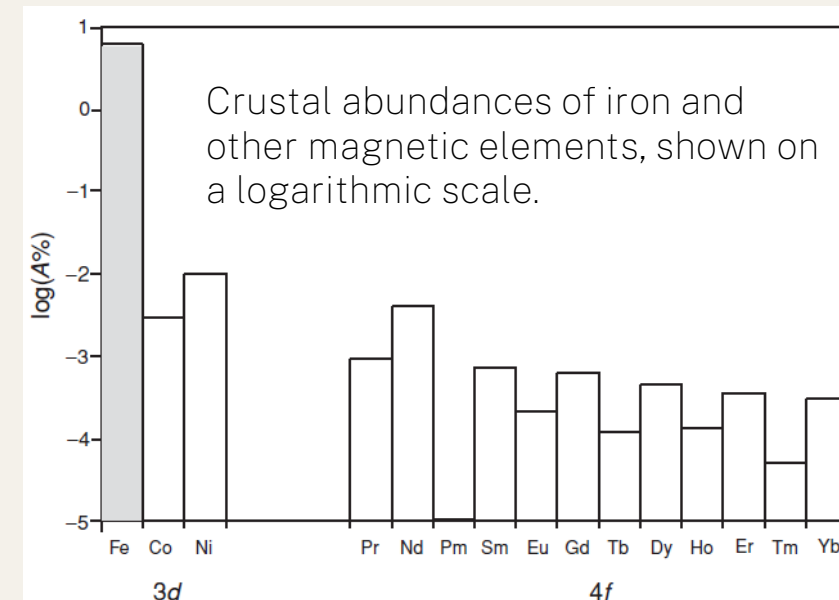
By attempting to answer Jeffreys' question we hope to present a relevant introduction to the theory and nature of magnetism in sediment and rocks.

What Makes Paleomagnetism Possible?

Signal	Planetary Magnetic Field (>3 Ga)
Recording Media	Fe is 4th most abundant element in crust (5% by wt in crust) • Fe readily forms oxides/sulfides, some are magnetic minerals • Fe has the property of permanent magnetism • Fe-oxides are common accessory minerals in rocks, sediments, soils (<1% vol)
Recording Processes	Geomagnetic field can be recorded during geological processes

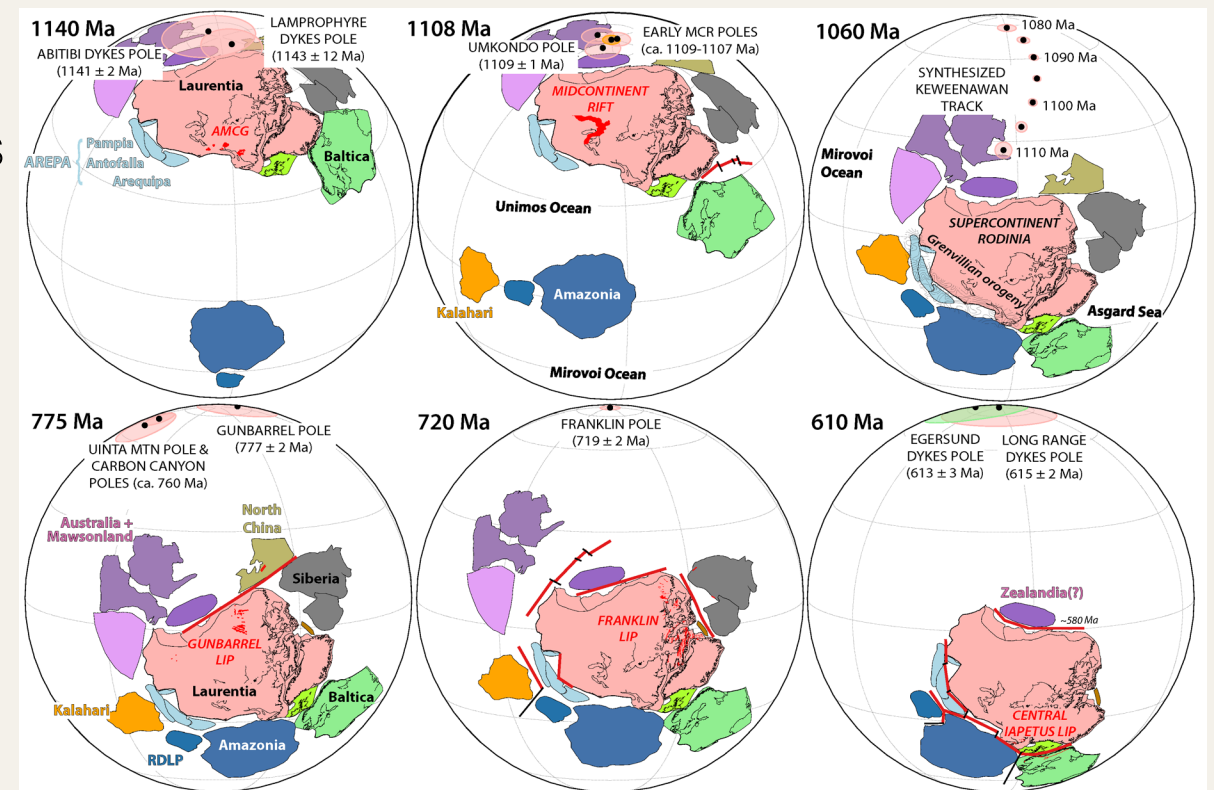
Iron (Fe^{2+} , Fe^{3+}) is most abundant magnetic element in crust. It is 40 times as abundant as all other magnetic elements together

Abundances of Magnetic Ions in the Earth's Crust



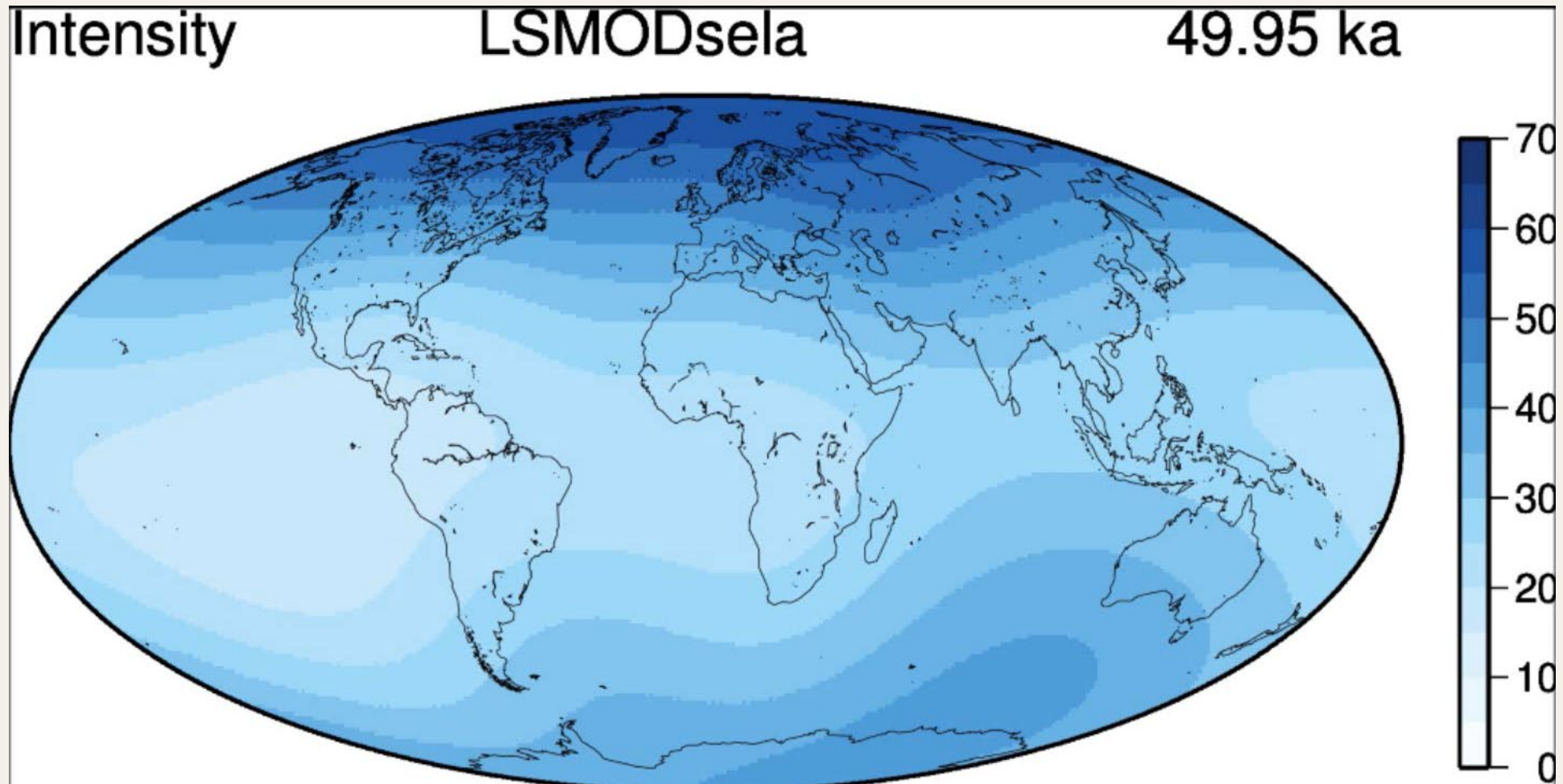
Why Study Paleomagnetism and Geomagnetism?

- Geologic and Geophysical History of lithospheric plate motions
- Probe of Deep Earth and Planetary Dynamics
- Geochronology and Magnetostratigraphy
- Environmental Magnetism & Paleoclimatic Reconstructions
- Geomicrobiology and Magnetofossils
- Magnetism of Extraterrestrial Materials
- Geohealth



Swanson-Hysell, N. L. (2021)
The Precambrian paleogeography of Laurentia

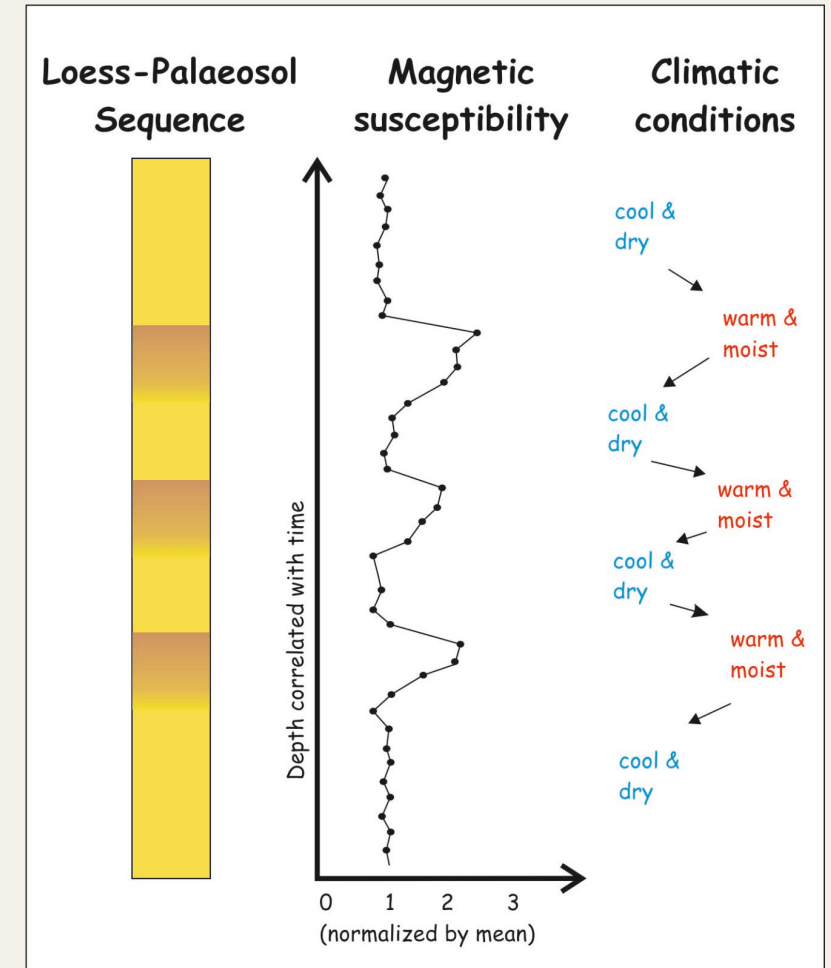
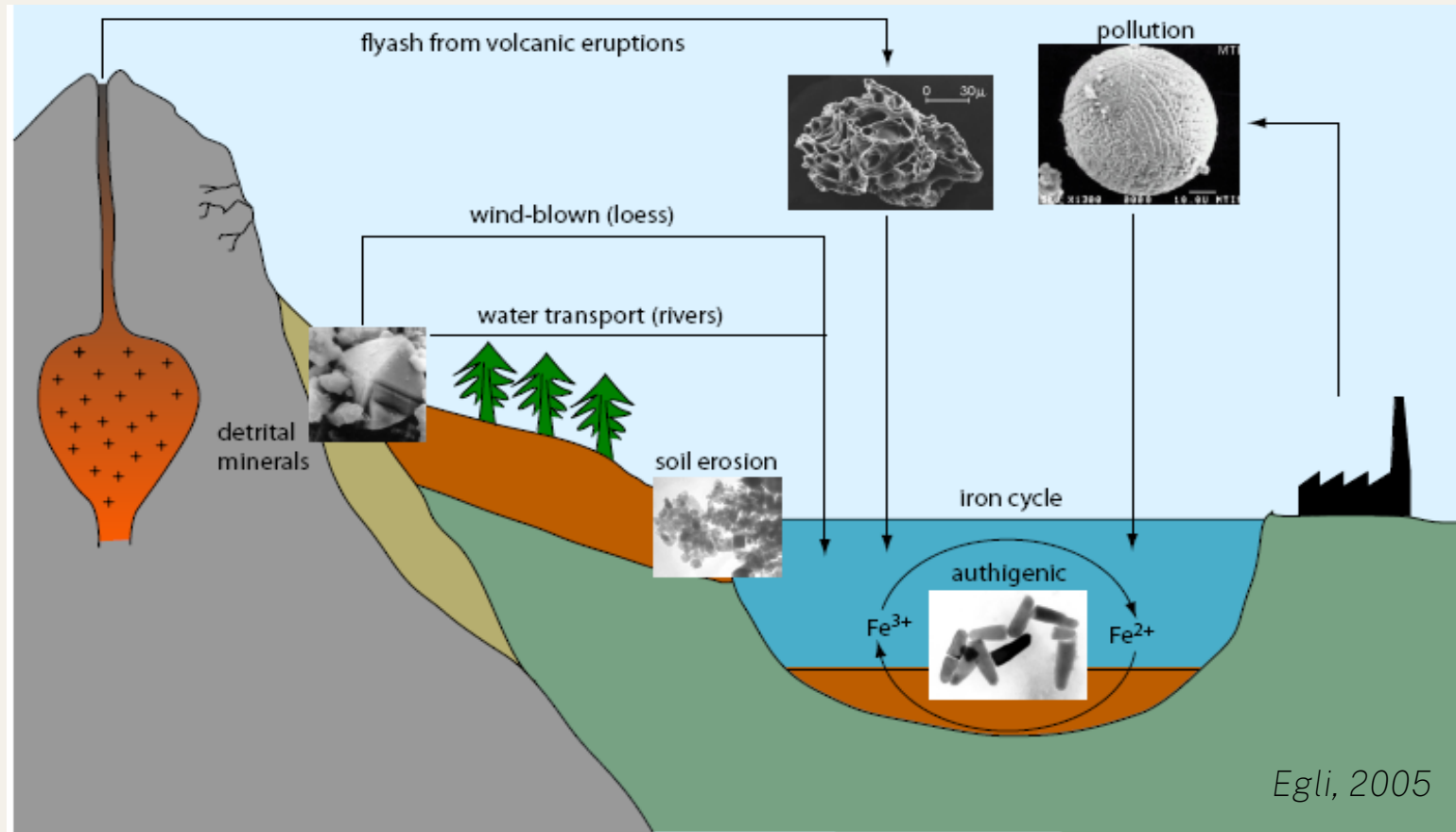
Past Magnetic Fields



Laschamp Excursion Intensity Variations (μT) (Brown et al. 2018)

Environmental Magnetism

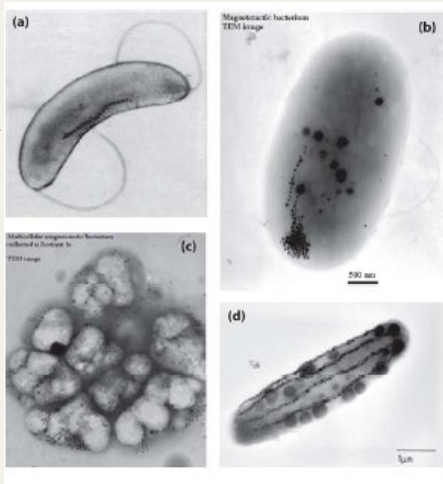
Paleoclimatic Reconstructions



Stephanie Scheidt -
<https://blogs.egu.eu/divisions/cl/files/2019/07/fig.-2-enviromagnetics.jpg>

Geomicrobiology and Magnetofossils

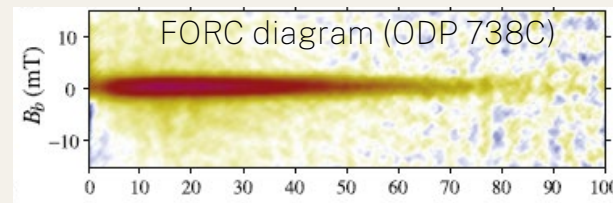
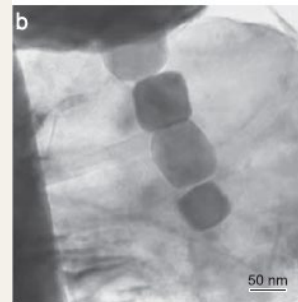
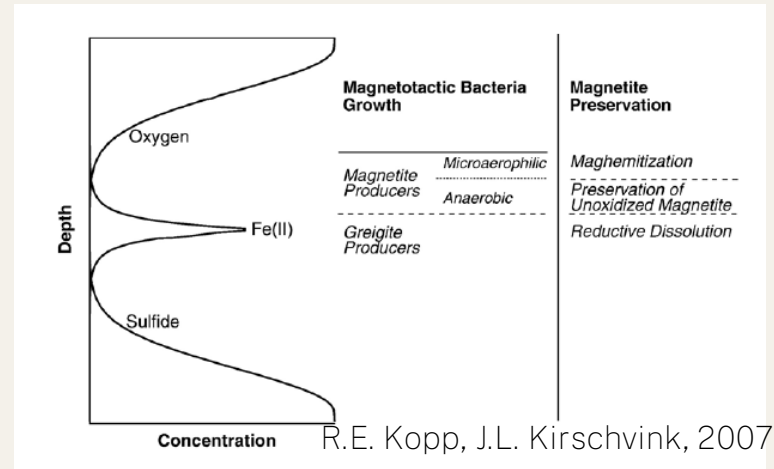
Favre and Schuler, 2008



Magnetotactic bacteria synthesize chains of magnetic nanoparticles that help them navigate in the geomagnetic field

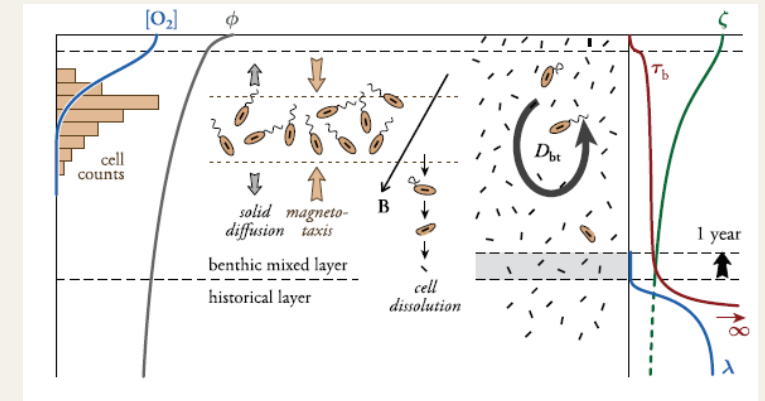


Superpara magnetic particles produced by iron-reducing bacteria

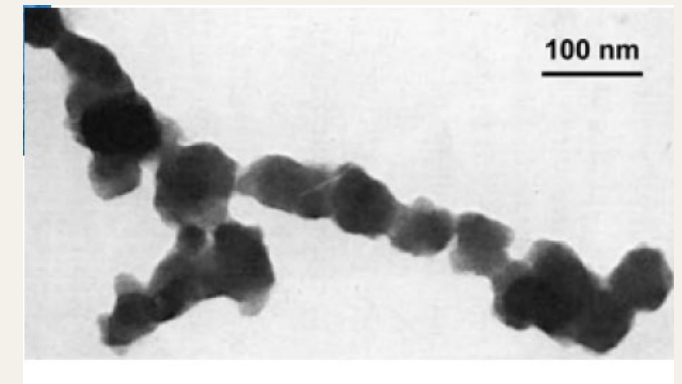


Magnetofossils from ODP Hole 738C sediments (southern Ocean, 56 Ma, Larrasoana et al. 2012)

Magnetofossil NRM acquisition model.

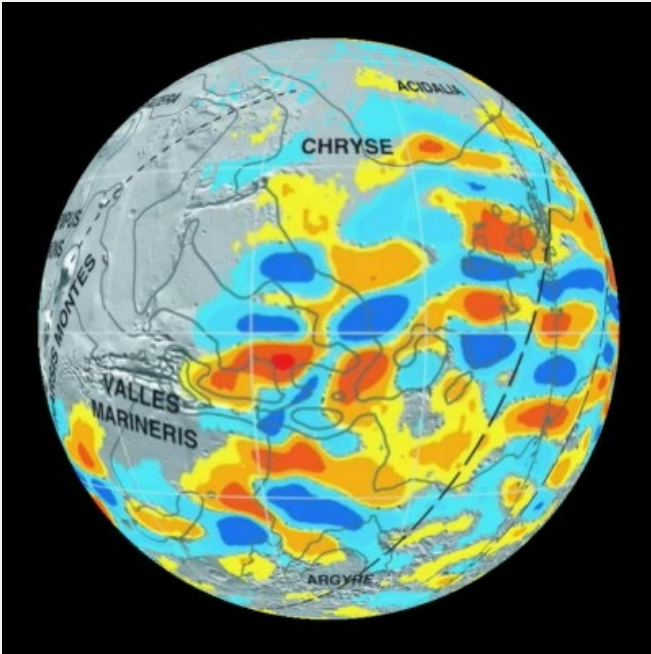


Mao et al, G³, 15, 2014



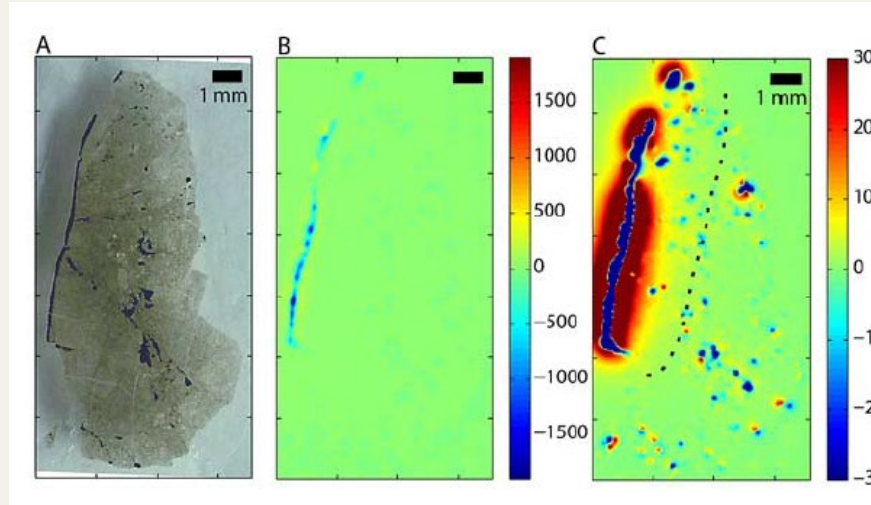
Possible oldest magnetofossils
~2 Gyr stromatolitic chert
(Chang et al., 1989)

Magnetism of Extraterrestrial Materials



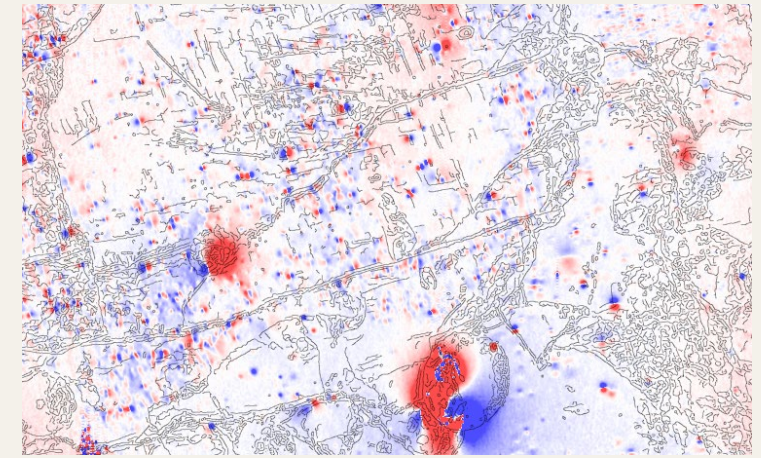
Crustal Magnetic Field of Mars

<https://sos.noaa.gov/catalog/datasets/mars-magnetic-map/>



Martian meteorite ALH 84001

Map of magnetic fields (in nT) measured with the SQUID microscope (Weiss et al., 2008)



Mercury-like meteorite NWA-1595A

QDM image of iron sources in lamellae and larger sulphides (Maxwell Brown)

Geohealth

Efficacy of green infrastructure in reducing exposure to local, traffic-related sources of airborne particulate matter (PM)

H.A. Sheikh^{a,*}, B.A. Maher^b, A.W. Woods^a, P.Y. Tung^{a,c}, R.J. Harrison^a

^a Department of Earth Sciences, University of Cambridge, Downing Site, CB2 3EQ, UK

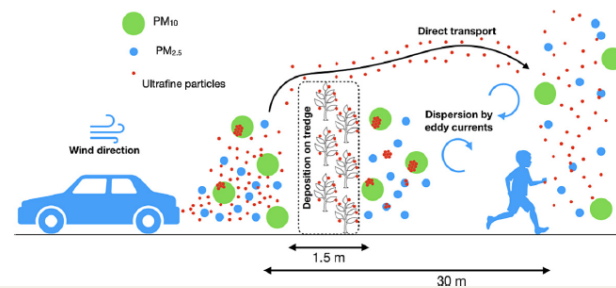
^b Center for Environmental Magnetism and Palaeomagnetism, University of Lancaster, LA1 4YQ, UK

^c Department of Materials Science and Metallurgy, University of Cambridge, CB3 0FS, UK

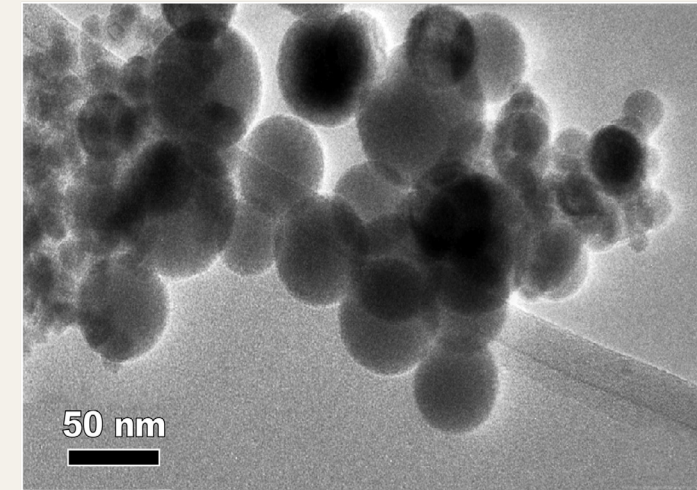
HIGHLIGHTS

- Magnetic PM loading is dominated by superparamagnetic (<30 nm) Fe-oxide nanoparticles.
- Tredge leaves capture an estimated 23 % of local, traffic-derived PM₁₀ emissions over two-day exposure period.
- Deposition and dispersion processes at the tredge reduce exposure to local, traffic-derived PM pollution.

GRAPHICAL ABSTRACT



TEM image of magnetite nanoparticles captured from the exhaust plume of a diesel engine.



Barbara A. Maher et al. PNAS 2016;113:39:10797-10801

Pollution

These particles are often enriched in nickel, platinum, cobalt, and copper.

www.pnas.org/cgi/doi/10.1073/pnas.1605941113



Magnetite pollution nanoparticles in the human brain

Barbara A. Maher^{a,1}, Imad A. M. Ahmed^b, Vassil Karloukovski^a, Donald A. MacLaren^c, Penelope G. Foulds^d, David Allsop^d, David M. A. Mann^e, Ricardo Torres-Jardón^f, and Lilian Calderon-Garciduenas^{g,h}

“We used magnetic analyses and electron microscopy to identify the abundant presence in the brain of magnetite nanoparticles that are consistent with high-temperature formation, suggesting, therefore, an external, not internal, source.”

Textbooks

